THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · EASY LEVEL UNIT TEST SET 2

Chapters 1–4: **Chemical Reactions and Equations, Acids, Bases and Salts, Metals and Non-metals, Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The valency of carbon is: 1

(a) 4 **(b)** 6 **(c)** 2 **(d)** 3

2. $2\,H_2O \longrightarrow 2\,H_2 + O_2$ is a: 1

(a) neutralisation reaction **(b)** displacement reaction
(c) combination reaction **(d)** decomposition reaction

3. A solution with pH 3 is: 1

(a) basic **(b)** acidic **(c)** neutral **(d)** salty

4. The metal most reactive among the following is: 1

(a) potassium **(b)** copper **(c)** silver **(d)** zinc

5. A double bond is present in: 1

(a) methane **(b)** propane **(c)** ethane **(d)** ethene

6. The colour of barium sulphate precipitate is: 1

(a) yellow **(b)** black **(c)** white **(d)** blue

7. Acid + metal gives salt and: 1

(a) carbon dioxide **(b)** hydrogen gas **(c)** water only **(d)** oxygen gas

8. The ore of aluminium is: 1

(a) zinc blende **(b)** haematite **(c)** cinnabar **(d)** bauxite

9. Which allotrope of carbon conducts electricity? 1

(a) fullerene **(b)** diamond **(c)** graphite **(d)** none

10. The chemical formula of quick lime is: 1

(a) $Ca(OH)_2$ **(b)** CaO **(c)** $CaCl_2$ **(d)** $CaCO_3$

11. Blue litmus paper turns red in: 1

(a) acid **(b)** salt solution **(c)** base **(d)** water

12. The shine of a metal is called: 1

(a) ductility **(b)** sonority **(c)** malleability **(d)** lustre

13. The simplest hydrocarbon is: 1

(a) ethane **(b)** methane **(c)** ethyne **(d)** ethene

14. $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ is a: 1

(a) displacement reaction **(b)** combination reaction
(c) double displacement reaction **(d)** decomposition reaction

15. A solution with pH 12 is: 1

(a) basic **(b)** acidic **(c)** neutral **(d)** none

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. What is the colour of phenolphthalein in 2
(i) acid
(ii) base?

17. Name the metals present in 2
(i) brass
(ii) bronze.

18. Define isomers. 2

19. What is corrosion? Name the product formed on iron. 2

20. Write the word equation for the reaction of an acid with a base. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. State three properties of carbon compounds. 3

22. An iron nail is kept in copper sulphate solution. State three observations. 3

23. What is the effect of acid rain on aquatic life? When is rain called acid rain? 3

24. State three uses of metals (e.g. copper, aluminium, iron). 3

OR

Name the products of ethanoic acid and sodium hydroxide reaction and write the equation. 3

25. Balance the equation $N_2 + H_2 \longrightarrow NH_3$. 3

OR

Write the products of the reaction of dilute HCl with magnesium. Give the balanced equation. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. What is corrosion? State the conditions necessary for rusting, name rust and give three ways to prevent it. 5

OR

Write the preparation, formula and two uses of 5
(a) washing soda and
(b) plaster of Paris.

27. What is an alloy? Write the composition and uses of brass, bronze, solder, stainless steel and amalgam. 5

OR

Describe reactions of ethanoic acid with sodium hydroxide, sodium carbonate and ethanol. 5

END OF QUESTION PAPER **Total: 50 Marks**

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